

In the previous topics, we have discussed that the concept of the random experiment leads to the notation of a sample space. The assignment & computational of probabilities of events were studied in detail. In many experiments, we are interested not in knowing which of the outcomes has occurred, but <sup>in</sup> the number associated with them.

For example, when 'n' coins are tossed, one may be interested in knowing the number of heads obtained. When a pair of dice are tossed, one may seek information about the sum of points. Thus, we associated a real number with each outcome of an experiment.

In other words, we are considering a function whose domain is the set of possible outcomes, & whose range is a subset of the set of reals. Such a function is called a random variable.

Def:- A random variable (r.v.) is a function  $X(\omega)$  with domain  $S$  & range  $(-\infty, \infty)$  such that for every real number 'a', the event  $\boxed{[\omega: X(\omega) \leq a] \in B.}$

Remark-① The function  $X(\omega)$  is measurable real function on  $(S, B)$ .

### Some Theorems on Random Variables

- Consider the probability space  $(S, B, P)$
1. A function  $X(\omega)$  from  $S$  to  $R(-\infty, \infty)$  is a r.v. iff for real  $a$ ,  $\{\omega: X(\omega) \leq a\} \in B$
  2. If  $X_1$  &  $X_2$  are random variables &  $C$  is constant then  $CX_1, X_1 + X_2, X_1 X_2$  are also random variables.
  3. If  $X$  is a random variable then
    - (i)  $\frac{1}{X}$ , where  $(\frac{1}{X})(\omega) = \infty$  if  $X(\omega) = 0$
    - (ii)  $X_+(\omega) = \max\{0, X(\omega)\}$
    - (iii)  $X_-(\omega) = -\min\{0, X(\omega)\}$ ,
    - (iv)  $|X|$  are random variables.

Remark ② If  $x$  is a real number, the set of all  $\omega$  in  $S$  such that

$$X(\omega) = x$$

denoted briefly  $X = x$

$$\text{Thus } P(X=x) = P\{\omega: X(\omega) = x\}$$

$$\text{Similarly } P(X \leq a) = P\{\omega: X(\omega) \in (-\infty, a]\}$$

$$\& P(a < X \leq b) = P\{\omega: X(\omega) \in (a, b]\}$$

Analogous meanings are given to  $P(X=a \text{ or } X=b) = P\{(X=a) \cup (X=b)\}$

$$\Rightarrow P(X=a \& X=b) = P\{(X=a) \cap (X=b)\}$$

eg-① if  $E$  consists of two tosses, the rv which is the number of heads (0, 1 or 2)

|                |    |  |    |    |    |
|----------------|----|--|----|----|----|
|                | HH |  | HT | TH | TT |
| Outcome :      |    |  |    |    |    |
| Value of $X$ : | 2  |  | 1  | 1  | 0  |

Thus to each outcome  $\omega$ , there corresponds a real number  $X(\omega)$ .

② An experiment consists of rolling a die & reading the number of points on the upturned face. The most natural r.v.  $X$  to consider is

$$X(\omega) = \omega; \omega = 1, 2, \dots, 6$$

③ If we are interested in knowing whether the number of points is even or odd, we consider a r.v.  $Y$  defined as follows -

$$Y(\omega) = \begin{cases} 0, & \text{if } \omega = \text{even} \\ 1, & \text{if } \omega = \text{odd} \end{cases}$$

(4) If  $X_1$  &  $X_2$  are random variables, then

(21)

(i)  $\max[X_1, X_2]$  &

(ii)  $\min[X_1, X_2]$

are also random variables.

(5) If  $X$  is a r.v. &  $f(\cdot)$  is a continuous function, then  $f(X)$  is a r.v.

(6) If  $X$  is a r.v. &  $f(\cdot)$  is an ~~continuous~~ <sup>increasing</sup> function, then  $f(X)$  is a r.v.

### Distribution Function (d.f.)

Def<sup>n</sup>: Let  $X$  be a random variable. The function  $F$  defined for all real  $x$  by —

$$F(x) = P\{X \leq x\} = P\{\omega: X(\omega) \leq x\}, \quad -\infty < x < \infty$$

is called the <sup>function</sup> distribution function (d.f.) of the r.v. ( $X$ ).  
Remark: — A distribution is also called the cumulative distribution function. Sometimes, the notation  $F_X(x)$  is used to emphasize the fact that the distribution function is associated with the particular random variable  $X$ . Clearly, the domain of the distribution function is  $(-\infty, \infty)$  & its range is  $[0, 1]$ .

### Properties

(1) If  $F$  is the d.f. of the r.v.  $X$  & if  $a < b$ , then

$$P(a < X \leq b) = F(b) - F(a)$$

(2) If  $F$  is d.f. of one-dimensional r.v.  $X$ , then

(i)  $0 \leq F(x) \leq 1$

(ii)  $F(x) \leq F(y)$  if  $x < y$

In other words, all d.f.<sup>n</sup>  $f^n$  are monotonically non-decreasing & lie between 0 & 1.

(3) If  $F$  is d.f. of one-dimensional r.v.  $X$ , then

$$F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0 \quad \& \quad F(\infty) = \lim_{x \rightarrow \infty} F(x) = 1$$



### Proof - Property-1

The event ' $a < X \leq b$ ' & ' $X \leq a$ ' are disjoint & their union is the event ' $X \leq b$ '. Hence by addition theorem of probability -

$$P(a < X \leq b) + P(X \leq a) = P(X \leq b)$$

$$\Rightarrow P(a < X \leq b) = P(X \leq b) - P(X \leq a)$$
$$= F(b) - F(a)$$

Corollary-1  $P(a \leq X \leq b) = P\{(X=a) \cup (a < X \leq b)\}$

$$= P(X=a) + P(a < X \leq b)$$
$$= P(X=a) + [F(b) - F(a)]$$

Similarly,

$$P(a < X < b) = P(a < X \leq b) - P(X=b)$$
$$= F(b) - F(a) - P(X=b)$$

$$P(a \leq X < b) = P(a < X < b) + P(X=a)$$
$$= F(b) - F(a) - P(X=b) + P(X=a)$$

Remark When  $P(X=a) = 0$  &  $P(X=b) = 0$ , all four events  $a \leq X \leq b$ ,  $a < X < b$ ,  $a \leq X < b$  &  $a < X \leq b$  have the same probability  $F(b) - F(a)$

## Discrete Random Variable

A variable which can assume only a countable number of real values & for which the value which the variable takes depends on chance, is called a discrete random variable.

In other words, a real valued function defined on a discrete sample space is called a discrete random variable.

eg- marks obtained in a test, number of accidents per month, number of telephone call per unit time, number of success in trials, & so on.

## Probability Mass Function

If  $X$  is a one-dimensional discrete random variable taking at most a countably infinite number of values  $x_1, x_2, \dots$  then its probabilities behaviour at each point is described by a function called the probability mass function.

Def<sup>n</sup>: - If  $X$  is a discrete random variable with distinct values  $x_1, x_2, \dots, x_n, \dots$  then the function  $p(x)$  defined as:

$$p_X(x) = \begin{cases} p(X=x_i) = p_i, & \text{if } x=x_i \\ 0 & \text{if } x \neq x_i; i=1,2,\dots \end{cases}$$

is called the probability mass function of r.v.  $X$ .

Remark ① The number  $p(x_i); i=1,2,\dots$  must satisfy the following conditions -

(i)  $p(x_i) \geq 0 \forall i$

(ii)  $\sum_{i=1}^{\infty} p(x_i) = 1$

② The set of values which  $X$  takes is called the spectrum of the r.v.  
eg. Toss of a coin,  $S = \{H, T\}$ . Let  $X$  be the r.v. defined as -

$$\begin{aligned} X(H) &= 1 \text{ i.e. } X=1 \text{ if Head occurs} \\ X(T) &= 0 \text{ i.e. } X=0 \text{ if Tail occurs} \end{aligned}$$

② The set of values which  $X$  takes is called the spectrum of the random variable.

The set of ordered pairs  $\{x_i, p(x_i); i=1, 2, \dots, n, \dots\}$  or  
 $\{(x_1, p_1), (x_2, p_2), \dots, (x_n, p_n), \dots\}$  specifies the prob.  
distribution of the r.v.  $X$ .

# Discrete Distribution Function

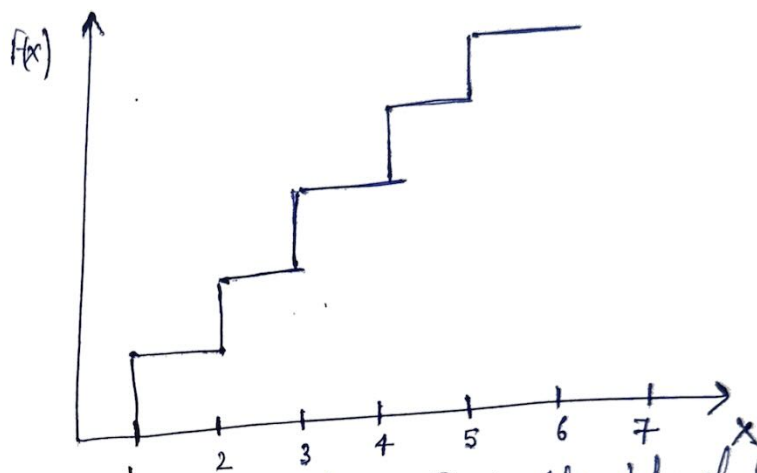
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In this case, there are a countable number of points  $x_1, x_2, x_3, \dots$  & number

$$p_i \geq 0, \sum_{i=1}^{\infty} p_i = 1 \text{ s.t.}$$

$$F(x) = \sum_{i: x_i \leq x} p_i$$

eg- if  $x_i$  is just the integer  $i$ , so that  $p(X=i) = p_i, i=1, 2, \dots, 3, \dots$  then  $F(x)$  is a 'step function' having jump  $p_i$  at  $i$  & being constant between each pair of integers.



Theorem  $p(x_j) = P(X=x_j) = F(x_j) - F(x_{j-1})$ , where  $F$  is the d.f. of  $X$

Proof Let  $x_1 < x_2 < \dots$  we have

$$F(x_j) = P(X \leq x_j) = \sum_{i=1}^j P(X=x_i) = \sum_{i=1}^j p(x_i)$$

$$\& F(x_{j-1}) = P(X \leq x_{j-1}) = \sum_{i=1}^{j-1} p(x_i)$$

$$\therefore F(x_j) - F(x_{j-1}) = p(x_j)$$

Thus, given the distribution fn of discrete r.v., we can compute pmf.



# Continuous Random Variable

A random variable  $X$  is said to be continuous if it can take all possible values (fractional as well as integral) between certain limits. In other words, a random variable is said to be continuous when its different values cannot be put in 1-1 correspondence with a set of positive integers.

A continuous random variable is a variable that can be measured to any desired degree of accuracy.  
eg- age, height, weight, etc.

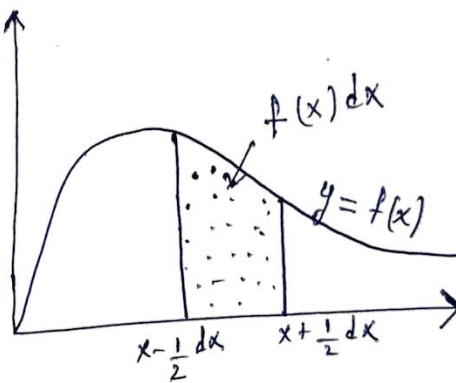
## Probability Density Function

Consider the small interval  $(x, x+dx)$  of length  $dx$  round the point  $x$ . Let  $f(x)$  be any continuous fn of  $x$  so that  $f(x)dx$  represents the probability that  $X$  falls in the infinitesimal interval  $(x, x+dx)$ .

Symbolically,

$$P(x \leq X \leq x+dx) = f(x)dx$$

In the figure,  $f(x)dx$  represents the area bounded by the curve  $y=f(x)$ ,  $x$ -axis & ordinates at the points  $x$  &  $x+dx$ . The function  $f(x)$  so defined is known as probability density function or simply density function of random variable  $X$  & is usually abbreviated as pdf.



The expression  $f(x)dx$  usually written as  $dF(x)$ , is known as the probability differential & the curve  $y=f(x)$  is known as the pdf or simply probability curve.

Def<sup>n</sup> pdf  $f(x)$  of the r.v.  $X$  is defined as:

$$f(x) = \lim_{\delta x \rightarrow 0} \frac{P(x \leq X \leq x+\delta x)}{\delta x}$$



The probability for a variate value to lie in the interval  $dx$  is  $f(x) dx$  & hence the probability for a variate value to fall in the finite interval  $[a, b]$  is:

$$P(a \leq X \leq b) = \int_a^b f(x) \cdot dx$$

which represents area between the curve  $y = f(x)$  & the ordinates at  $x = a$  &  $x = b$ . Further, since total probability is unity, we have —

$$\int_a^b f(x) \cdot dx = 1, \text{ where } [a, b] \text{ is the range of the random variable } X.$$

The range of the variable may be finite or infinite.

The probability density function (pdf) of a random variable (r.v.)  $X$ , usually denoted by  $f_X(x)$  or simply  $f(x)$  has the following obvious properties —

(i)  $f(x) \geq 0$

(ii)  $\int_{-\infty}^{\infty} f(x) \cdot dx = 1$

(iii) The Probability  $P(E)$  given by —

$$P(E) = \int_E f(x) \cdot dx$$

is well defined for any event  $E$ .

Important Remark — In case of discrete random variable, the prob. at a point i.e.  $P(X=c)$  is not zero for some fixed  $c$ . However, in case of continuous random variables the probability at a point is always zero.

## Continuous Distribution Function

Def<sup>n</sup>:- If  $X$  is a continuous r. variable with the pdf  $f(x)$ , then the function -

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt, \quad -\infty < x < \infty$$

is called the distribution f<sup>n</sup> d.f. or ~~some~~ sometimes the cumulative distribution f<sup>n</sup> (cdf) of the r.v.  $X$ .

### Properties

1.  $0 \leq F(x) \leq 1, -\infty < x < \infty$
2. From analysis (Riemann integral), we know that  
$$F'(x) = \frac{d}{dx} F(x) = f(x) \geq 0$$
  
 $\Rightarrow F(x)$  is non-decreasing f<sup>n</sup> of  $x$ .
3.  $F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = \lim_{x \rightarrow -\infty} \int_{-\infty}^x f(x) dx = \int_{-\infty}^{-\infty} f(x) dx = 0$   
 $\times F(\infty) = \lim_{x \rightarrow \infty} F(x) = \lim_{x \rightarrow \infty} \int_{-\infty}^x f(x) dx = \int_{-\infty}^{\infty} f(x) dx = 1$

4.  $\Rightarrow 0 \leq F(x) \leq 1$
5.  $F(x)$  is a continuous f<sup>n</sup> of  $x$  on the right.
6. The discontinuities of  $F(x)$  are at the most countable.
7. It may be noted that

$$P(a \leq X \leq b) = \int_a^b f(x) \cdot dx = \int_{-\infty}^b f(x) \cdot dx - \int_{-\infty}^a f(x) \cdot dx$$

$$= P(X \leq b) - P(X \leq a)$$

$$= F(b) - F(a)$$

$$\text{Similarly, } P(a < X < b) = P(a < X \leq b) = P(a \leq X < b) = \int_a^b f(t) dt$$

$$7. F'(x) = \frac{d}{dx} F(x) = f(x) \Rightarrow dF(x) = f(x) dx$$

# Various Measures of Central Tendency, Dispersion, Skewness & Kurtosis for Continuous Probability Distributions

Let  $f_X(x)$  or  $f(x)$  be the pdf of r.v.  $X$ , where  $X$  is defined from  $a$  to  $b$ . Then -

(i) Arithmetic Mean  $= \int_a^b x \cdot f(x) dx$

(ii) Harmonic Mean  $: \frac{1}{H} = \int_a^b \frac{1}{x} \cdot f(x) dx$

(iii) Geometric Mean  $: \log G = \int_a^b \log x \cdot f(x) dx$

(iv)  $\mu_r'$  (about origin)  $= \int_a^b x^r \cdot f(x) dx$

$\mu_r'$  (about the point  $x=A$ )  $= \int_a^b (x-A)^r \cdot f(x) dx$

&  $\mu_r'$  (about mean)  $= \int_a^b (x - \text{mean})^r \cdot f(x) dx$

In particular,  $\mu_1'$  (about origin) = Mean  $= \int_a^b x \cdot f(x) dx \leftarrow$

$\mu_2' = \int_a^b x^2 \cdot f(x) dx$

Hence  $\mu_2 = \mu_2' - \mu_1'^2$

$\mu_2 = \int_a^b x^2 \cdot f(x) dx - \left( \int_a^b x \cdot f(x) dx \right)^2 \leftarrow$

$\mu_3 = \mu_3' - 3\mu_2' \mu_1' + 2\mu_1'^3$

$\mu_4 = \mu_4' - 4\mu_3' \mu_1' + 6\mu_2' \mu_1'^2 - 3\mu_1'^4$

& hence  $\beta_1$  &  $\beta_2$  can be computed

(v) Median  $\rightarrow$  Median is the point which divides the entire dist<sup>n</sup> in two equal parts. In case of continuous dist<sup>n</sup>, median is the point which divides the total area into two equal parts.

thus if  $M$  is the median, then

$$\int_a^M f(x) \cdot dx = \int_M^b f(x) \cdot dx = \frac{1}{2}$$



(vi) Mean Deviation :- M.D. about the mean  $\mu$ , is given by

$$MD = \int_a^b |x - \text{mean}| f(x) dx$$

In general, mean deviation about an average  $A$  is given by:

$$\text{M.D. about } A = \int_a^b |x - A| f(x) dx$$

(vii) Quartiles & Deciles :-  $Q_1$  &  $Q_3$  are given by the eqn -

$$\int_a^{Q_1} f(x) dx = \frac{1}{4} \quad \& \quad \int_a^{Q_3} f(x) dx = \frac{3}{4}$$

$D_i$ ,  $i$ th decile is given by

$$\int_a^{D_i} f(x) dx = \frac{i}{10}, \quad i = 1, 2, \dots, 9$$

(viii) Mode is the value of  $x$  for which  $f(x)$  is maximum.

$$f'(x) = 0$$

$$\& \quad f''(x) < 0$$

provided it lies on  $[a, b]$